THE COHOMOLOGICAL DIMENSION OF A FIELD EXTENSION IS BOUNDED BY THE COHOMOLOGICAL DIMENSION OF THE BASE FIELD PLUS THE TRANSCENDENCE DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the tower reduction to transcendence degree one, the presentation of the field as a filtered colimit of finite type subalgebras with the resulting reduction, and the final curve bound.

Editorial note. The label lemma-cd-field-extension names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source etale-cohomology.tex –label lemma-cd-field-extension –start 19672 –end 19689 –local-id 3561 returned RESULT=OK (L3561 free, occupancy FREE, public interval clear). Candidate taken from lane-s4/halfB-priority.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Cohomological dimension of a filtered colimit of rings. Same chapter, the lemma bounding the cohomological dimension of a filtered colimit of algebras by the dimensions of the constituents. Supplies the reduction from L to its finite type K-subalgebras.
- Cohomological dimension of a curve over a field. Same chapter, the lemma bounding the cohomological dimension of the function field of a curve over a field of given cohomological dimension. Supplies the bound cd(A_i) <= 1 + cd(K) for a finite type K-subalgebra of transcendence degree one, which finishes the proof.

Source reference. Original reference 0F0S: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19651–19651. Prerequisite pointer invoked at proof line(s) 19688; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0F0R: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19633–19633. Prerequisite pointer invoked at proof

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

line(s) 19686; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let L/K be a field extension. Then we have $cd(L) \leq cd(K) + trdeg_K(L)$.*

Proof. If $\mathrm{trdeg}_K(L) = \infty$, then this is clear. If not then we can find a sequence of extensions $L = L_r/L_{r-1}/\ldots/L_1/L_0 = K$ such that $\mathrm{trdeg}_{L_i}(L_{i+1}) = 1$ and $r = \mathrm{trdeg}_K(L)$. Hence it suffices to prove the lemma in the case that $r = 1$. In this case we can write $L = \mathrm{colim}\, A_i$ as a filtered colimit of its finite type K-subalgebras. By Lemma 0F0R it suffices to prove that $\mathrm{cd}(A_i) \leq 1 + \mathrm{cd}(K)$. This follows from Lemma 0F0S. □